Chloe Rose Elia

McKinney, Texas
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Academic Profile

Undergraduate student at the University of Alaska Fairbanks pursuing a Bachelor of Science in Biological Sciences with a concentration in Biomedical Science, while concurrently completing an Associate of Science in General Science.

Interested in biotechnology, biomedical research, aquatic health, and biological data science. Particularly drawn to interdisciplinary research connecting biological sciences, computational tools, and engineering. Long-term academic interests include bioinformatics, quantitative biological analysis, and emerging applications of technology in biomedical and aquatic research.

Education

University of Alaska Fairbanks

Bachelor of Science in Biological Sciences
Concentration: Biomedical Science
Expected May 2029

Associate of Science in General Science
Expected Spring 2027

Both degree programs pursued concurrently.

Research Experience

Undergraduate Research — Yukon River Chinook Salmon Health
University of Alaska Fairbanks · 2026–Present

- Contributing to a research project investigating the health of returning adult Yukon River Chinook salmon.
- Investigating the relationship between visible cardiac spots and histological evidence of *Ichthyophonus* infection.
- Developing experience with digital histology and quantitative image analysis using QuPath.
- Exploring measurements of parasite size, inflammatory lesions, and affected cardiac tissue.
- Applying computational image-analysis tools to biological research questions.

Research Interests

- Biotechnology and biomedical science
- Bioinformatics and biological data science
- Digital histology and quantitative image analysis
- Aquatic animal health and disease
- Molecular and cellular biology
- Marine and freshwater biology
- Microscopy and biological imaging
- Biomedical engineering and emerging technologies

Technical Skills and Research Tools

Research Software

- QuPath — introductory experience with digital histology and image analysis
- R — academic coursework and introductory statistical applications

Scientific and Analytical Interests

- Histological image analysis
- Quantitative biological measurements
- Biological data interpretation
- Scientific literature review
- Research documentation and scientific writing

Mechanical and Technical Experience

- Marine mechanical maintenance and troubleshooting
- Boat maintenance and repair
- Practical mechanical problem-solving

Professional Experience

Independent Marine Mechanic
Independent · 2018–2022

- Performed boat maintenance and mechanical repairs.
- Troubleshooting mechanical issues and supporting vessel upkeep.
- Developed practical mechanical and technical problem-solving experience.

Bartender and Server
Venezia Trattoria · 2026

Volunteer Experience

Volunteer Boat Crew / Captain
Mexico · 2020–2024

- Participated in volunteer boating activities and marine-related assistance.
- Contributed to vessel operations and practical support during volunteer activities.

Additional Information

- Academic awards and scholarships
- Research presentations
- Certifications and professional memberships

References

Available upon request.